Identifying Causal Effects in Information Provision Experiments

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Information provision experiments are an increasingly popular tool to identify how beliefs causally affect decision-making and behavior. In a simple Bayesian model of belief formation via costly information acquisition, people form precise beliefs when these beliefs are important for their decision-making. The precision of prior beliefs controls how much their beliefs shift when they are shown new information (i.e., the strength of the first stage). Since two-stage least squares (TSLS) targets a weighted average with weights proportional to the strength of the first stage, TSLS will overweight individuals with smaller causal effects and underweight those with larger effects, thus understating the average partial effect of beliefs on behavior. In experimental designs where all participants are exposed to new information, Bayesian updating implies that a control function can be used to identify the (unweighted) average partial effect. I apply this estimator to a recent study of the effects of beliefs about the gender wage gap on support for public policies (Settele, 2022) and find the average partial effect is 40% larger than the comparable TSLS estimate. This difference can be explained by the fact that the effects of beliefs are close to zero for people who update their beliefs the most and receive the most weight in TSLS specifications.

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Nobody knows everything. Important decisions, like how much schooling to acquire, depend on beliefs about the return to education and not directly on the return to education itself. While it may be difficult to experimentally vary the returns to education, it is easy to generate variation in beliefs by experimentally providing people with accurate new information (Jensen, 2010). For these reasons, information provision experiments have become a popular experimental tool to identify how beliefs causally affect decision-making and behavior.[1] In the first stage of these experiments, researchers experimentally vary the information (or "signal") shown to participants. This generates variation in beliefs that is used to study the effects of beliefs on behavior.

Two-stage least square (TSLS) regressions where random assignment to treatment is used to construct an instrument for beliefs are often used to estimate causal effects of beliefs. When beliefs have heterogeneous effects on behaviors, TSLS targets a weighted average of these heterogeneous effects with weights proportional to the variation induced in the first stage. TSLS puts the most weight on people who update their beliefs the most in response to new information. Coefficients in common specifications will thus differ from the average effect of beliefs to the extent that the variation in the first stage (i.e., how much people update beliefs in response to new information) is correlated with the heterogeneous effects of beliefs.

A simple Bayesian model of belief updating is a workhorse in the applied literature (for example Balla-Elliott et al., 2022; Cavallo et al., 2017; Cullen et al., 2022; Cullen and Perez-Truglia, 2022; Fuster et al., 2022). This model implies that a wide range of TSLS specifications common to the literature weight individuals proportionally to their Bayesian learning rate, which is large when prior beliefs are imprecise. In a model where beliefs are formed via costly information acquisition, people have imprecise beliefs when these beliefs are not important for their decision-making. This dynamic may explain why some experiments have found small or insignificant effects of beliefs on behavior, despite having treatments that have a large effect on beliefs (e.g. Kuziemko et al., 2015).

I propose an alternative estimator that uses the structure of the Bayesian learning model to directly estimate the average partial effect (APE) and confirm the theoretical prediction about the direction of the TSLS bias. This control function strategy is available in so-called "active" experimental designs, where study participants are randomly assigned to see a "high" or "low" signal (e.g. Akesson et al., 2022; Bottan and Perez-Truglia, 2022b; Roth et al., 2022).

[1] Haaland et al. (2023) provide an excellent recent review of this literature.

I apply this estimator to a recent study of the effect of beliefs about the gender wage gap on demand for public policy intervention (Settele, 2022). In this exercise, the average partial effect of beliefs on demand for public policy is roughly 40% larger than the corresponding TSLS estimate. I use the observed beliefs before and after the information treatment to order people by the amount that they update in response to new information, conditional on their prior beliefs. Average effects are close to zero for the 20% of people who update their beliefs the most, but are nearly twice as large as the original TSLS estimate among the remaining 80%. While these conditional average partial effects are less precisely estimated, they are broadly consistent with the theoretical predictions. The control function estimator thus provides a natural way to empirically analyze the key relationship between the causal effects of beliefs and the responsiveness of these beliefs to new information.

When an active experimental design is not possible, these results provide a framework for using theory that relates belief formation to the causal effects of interest to extrapolate from TSLS estimates to average effects in the population. The empirical results confirm the main qualitative prediction of the simple model of belief formation. In settings where this structure is reasonable, TSLS estimates will generally understate the effect of beliefs on behavior. I suggest how this model may be extended to include richer dynamics of belief formation, such as rational inattention (Fuster et al., 2022).

This paper studies and builds on the large and growing applied literature that uses information provision experiments to study the effects of beliefs on behavior and decision-making. I focus on experimental designs where the information treatment is quantitative, for example, "*12 percent of the US population are immigrants*" (Grigorieff et al., 2020; Hopkins et al., 2019) and not treatments that are qualitative, for example informational videos (Alesina et al., 2021; Dechezlepretre et al., 2023; Stantcheva, 2023).[2]

The remainder of this paper has the following structure. Section 1 presents the model framework and shows how the key parameter in a common linear belief updating model can be micro-founded in a normal-normal Bayesian updating model. Section 2 introduces a simple model of belief formation in equilibrium (i.e., before the experiment). This simple model of belief formation via costly information acquisition makes a strong prediction: people with the largest causal effects of beliefs form the most

[2]Examples of the kind of quantitative information presented in these experiments are government statistics (Bottan and Perez-Truglia, 2022a; Kuziemko et al., 2015; Roth et al., 2022; Settele, 2022), previous responses from other participants (Balla-Elliott et al., 2022; Bursztyn et al., 2020; Coibion et al., 2021) or other institutional sources (Cullen et al., 2022; Cullen and Perez-Truglia, 2022). See Haaland et al. (2023) for a systematic treatment.

precise beliefs before the experiment and thus have the weakest first stage response to new information.

Section 3 surveys common TSLS specifications and makes clear the conditions under which they can be interpreted as a non-negatively weighted average of heterogeneous causal effects. The Bayesian updating model for the first stage generates intuitive expressions for the weights. Models that allow the Bayesian learning rate to depend on the causal effect of beliefs can be used to interpret the bias of TSLS estimates relative to the unweighted average effect.

Section 4 shows that Bayesian updating is sufficient to identify the average partial effect (APE) in experimental designs with an active control group. Section 5 applies this estimator to Settele (2022) and shows that the empirical results are broadly consistent with the theory.

I present some additional results in the Appendix. Appendix A provides a way to estimate causal parameters in a passive design even when prior beliefs cannot be observed and the signal does not shift beliefs monotonically. The proof of the main identification result is in Appendix B. Appendix C provides some additional results about the empirical replication.

It is important to note that I use the Bayesian learning structure throughout for clarity of exposition and because it is a common way that the first stage is analyzed in the empirical literature. However, the key results are robust to deviations from the strict Bayesian learning structure. Bayesian learning is an intuitive sufficient condition, but it is not necessary. The control function argument will use one implication of this model–scalar unobserved heterogeneity–but does not require a particular functional form. In some specifications, positive weights follow from the assumption only that people update (weakly) in the direction of the signal; in others, this needs to be strengthened slightly to the assumption that the posterior is a weighted average of the prior and the signal. Researchers who are comfortable making a Bayesian learning assumption can do so instead of directly assuming a less familiar condition like scalar unobserved heterogeneity.

1. Model Framework

This section defines the notation we will use throughout. We will work with a simple outcome equation that is linear in beliefs and allows for beliefs to have heterogeneous effects. Then, I will introduce notation for the different kinds of experiments we will

analyze.

There is now a robust literature that uses information provision experiments, and a range of designs now fall under this broad umbrella (Haaland et al., 2023). For example, some information provision experiments focus on how beliefs are formed in response to new information (Coibion et al., 2021; Fuster et al., 2022). There is also a set of papers with a more qualitative analysis of how beliefs affect behaviors. These papers use information treatments that are more qualitative – for example informational videos (Alesina et al., 2021; Dechezlepretre et al., 2023; Stantcheva, 2023).

The experimental designs that are the focus of this paper have two key features. First, they study how beliefs affect behavior and not only how beliefs respond to new information. Additionally, we will focus on experimental designs where the information treatment is quantitative, for example "*12 percent of the US population are immigrants*" (Grigorieff et al., 2020; Hopkins et al., 2019) and not treatments that are qualitative, for example *"[t]he chances of a poor kid staying poor as an adult are extremely large"* (Alesina et al., 2018).

These designs can be split into two groups: designs where all participants see some new information, but the particular signal is randomly assigned (active designs), and designs where only a subset of participants see new information and the remaining control group is not shown any signal (passive designs).

To analyze belief updating in these designs, we will follow the literature in using a model that is linear in the difference between the new signal and the prior. This linear first stage equation can be interpreted in a Bayesian model of belief updating. This will help exposition throughout by allowing us to express TSLS weights in terms of substantive features like the learning rate and the relative precision of prior beliefs.

1.1. Outcomes

The outcome equation is a linear model with heterogeneous coefficients on beliefs.

$$Y_i = \tau_i X_i + U_i \tag{1}$$

This is the canonical random coefficients model where Y_i is the outcome or behavior of interest, X_i is the belief, U_i is the structural error term, and τ_i is the partial effect of X_i on Y_i. We will assume that the beliefs X_i are endogenous $\left(\mathbb{E}\left[X_i U_i\right] \neq 0\right)$; Y_i can be arbitrarily affected by unobservables U_i. The only restriction here is that the causal effect for a given individual is linear.

The natural parameter of interest is the average partial effect (APE) of X_i on Y_i, denoted as $\mathbb{E}[\tau_i]$. This parameter has a simple interpretation as the average effect of beliefs on behaviors. In a recent example, Jäger et al. (2023) study the effect of workers' beliefs about their outside options on labor market outcomes. They pose the question "how much does a percentage point shift in beliefs about a workers' outside option causally shift workers' intended labor market behavior? " (p. 25). The average partial effect answers this question: on average, a one unit increase in beliefs causally shifts the outcome Y by $\mathbb{E}[\tau_i]$.

An important special case of the linear in parameters model is the log-log specification, which is often interpreted in applied work as an approximation to an elasticity.[3] The average coefficient on beliefs in the log-log model – sometimes called a behavioral elasticity (Haaland et al., 2023) – is thus a special case of an APE. All results about the APE therefore apply immediately to settings where the target parameter is a behavioral elasticity.

1.2. Information Treatment and Beliefs

We will denote treatments arms by Z_i and the *experimental design* by the set $\mathcal{Z}$ of possible treatments over which the researcher randomizes. The most common design in the applied literature is the *passive* design in which the researcher randomizes over a single treatment and a control group. I will use $z = 0$ to denote a treatment arm that is not shown any information so that the possible treatments in the passive design are given by $\mathcal{Z} = \{0, 1\}$. The *active* design is the case where the researcher randomizes over two (or more) possible information interventions. In this case, $\mathcal{Z} = \{1, 2\}$. In any design, treatment will be randomly assigned.

While the treatment Z_i will be randomly assigned, it is important to note that the realized signal $S_i(Z_i)$ can generally vary with individuals in a way that is not assumed to be independent of the structural unobservable U_i. For example, consider when $S_i(1)$ is a low estimate of home value and $S_i(2)$ is a high estimate of the home value as in Bottan and Perez-Truglia (2022a). The researcher will randomly assign an individual to see a high or low signal of their home value, but the (realized and counterfactual) signal values likely are not independent of the other model elements. Roth et al. (2022) explicitly model the two potential signals as noisy measures of the same object: increases in

[3]The log-log regression only point identifies the average elasticity $\mathbb{E}[\eta_i]$ in a random effects model of the form $Y = CX^{\eta_i}$. In more general models, the coefficient of interest does not identify the average elasticity but instead identifies a linear approximation in log space. Applied researchers are generally comfortable interpreting this coefficient as an approximation to an average elasticity.

unemployment during a recession. This makes clear that the difference between the signals is due to random noise due to difference in the data sources, but the level of the signals is not randomly assigned. The realized signal is only randomly assigned conditional on the potential signal values.

Formally, researchers randomly show participants the signal $S_i(z)$ that corresponds to their treatment arm z. Treatment is assigned randomly in the sense that Z_i is statistically independent of U_i, X_i^0, $S_i(\cdot)$, α_i, τ_i. With this in mind, we can write the potential outcomes for beliefs as

$$X_i(z) = \begin{cases} X_i^0 + \alpha_i\left(S_i(z) - X_i^0\right) & if\, z > 0 \\ X_i^0 & if\, z = 0 \end{cases} \tag{2}$$

We will use the structural equations (1) and (2) to study common empirical specifications.

1.3. Belief Potential Outcomes are Motivated by Bayesian Learning

The literature often microfounds this model in a Bayesian learning model with normally distributed beliefs (Balla-Elliott et al., 2022; Cullen and Perez-Truglia, 2022). Since this Bayesian updating form in the first stage will be helpful to guide exposition throughout, I will briefly show how these potential outcomes for beliefs are generated by a Bayesian learning model and relate the key coefficient α_i to model primitives.

Consider a group of individuals with uncertain prior beliefs. The subjective probability that the variable X_i takes the value x is given by the density of the normal distribution $\mathcal{N}\left(X_i^0, \sigma_{iX}^2\right)$. We thus interpret X_i^0 as the mean of the prior distribution. As shorthand, we will call X_i^0 the prior belief of an individual i, though in this Bayesian learning model it is more precisely the mean of the prior distribution.

People then observe a signal S_i about the quantity X. We model this signal as a draw from a distribution $\mathcal{N}(S_i^*, \sigma_{iS}^2)$. We interpret the variances of these distributions to reflect the subjective (inverse) precision of the prior and the signal. These variances are important only in their relative size. People for whom $\frac{\sigma_{iS}^2}{\sigma_{iX}^2}$ is large think their prior is more precise than the signal, whereas those for whom $\frac{\sigma_{iS}^2}{\sigma_{iX}^2}$ is small think that the signal is more precise than their prior.

The posterior is then a distribution

$$\mathcal{N}\left(\left(1-\alpha_i\right) X_i^0 + \alpha_i S_i, \frac{\sigma_{iS}^2 \sigma_{iX}^2}{\sigma_{iS}^2 + \sigma_{iX}^2}\right) \tag{3}$$

$$\text{where } \alpha_i \equiv \frac{\sigma_{iX}^2}{\sigma_{iS}^2 + \sigma_{iX}^2} \tag{4}$$

As with the prior, we will call the mean of this distribution the posterior X. Note that the mean of the posterior distribution is a weighted average of the prior X_i^0 and the signal S_i, where the weights are given by their relative precision.[4] We can also note that $\frac{\sigma_{iS}^2 \sigma_{iX}^2}{\sigma_{iS}^2 + \sigma_{iX}^2} < \sigma_{iX}^2$; intuitively, the posterior distribution is more precise than the prior distribution.[5] We can then relate the prior X_i^0, the signal S_i and the posterior X_i through the equation

$$X_i = (1-\alpha_i) X_i^0 + \alpha_i S_i \tag{5}$$

There is some direct empirical support for this Bayesian foundation of the linear updating model. For example, Roth et al. (2022) find that all of the belief updating in their study is driven by people who report being "very unsure", "unsure" or "somewhat unsure" and that people who are "sure" or "very sure" do not update their beliefs. Similarly Roth and Wohlfart (2020) find that people who are less confident in their prior beliefs update roughly twice as much as people who are more confident. This is part of a wide consensus in the literature that this model represents belief updating well, at least in the context of information provision experiments (Balla-Elliott et al., 2022; Cavallo et al., 2017; Cullen et al., 2022; Cullen and Perez-Truglia, 2022; Fuster et al., 2022; Giaccobasso et al., 2022).

2. Belief Formation Through Costly Information Acquisition

In this section, I use a simple model of endogenous information acquisition to show that the magnitude of the structural coefficient τ_i and the precision of beliefs will be positively correlated in equilibrium. Consider the following example that illustrates the key dynamic that causes people to (endogenously) have more precise beliefs when

[4] This derivation can be found in textbook treatments like Robert (2007) or Hoff (2009).

[5] There is experimental evidence that people randomized to the group receiving a signal report greater confidence in their posterior beliefs (Akesson et al., 2022; Cavallo et al., 2017).

these beliefs have a strong effect on their behavior.

Example: Valuing a Home. Alice and Bob are both listing their homes for sale and need to choose a price. They both have some belief about the market value of their home. Inaccurate beliefs are (potentially) costly through their effect on the list price.[6]

Suppose that Alice and Bob have different ways of choosing their list price. Alice benchmarks her list price exactly to the market value. Bob hates moving, so he lists his house at some very high price, irrespective of the market value. Thus, Alice's belief has a strong causal effect on her actions, whereas Bob's belief about the market value does not affect his actions at all.

If given the chance to obtain an informative signal, Alice will trade off the utility she gets from being more certain that she has correctly priced her house against the cost of the signal. If the cost of the signal is low enough, she will buy the signal until her beliefs are precise enough that the marginal gain in precision is less valuable than the cost of the signal. Since the price she sets depends on her beliefs about the market value, Alice will be well-informed about the market value.

Since Bob's beliefs do not affect his list price, his loss does not depend on his beliefs. Therefore, he will always decline the chance to buy a signal to update his beliefs, and his beliefs will remain centered around an idiosyncratic prior. Since the price he sets does not depend on his beliefs about the market value, Bob will remain poorly informed about it.

Now consider an information experiment with Bob and Alice in the sample. If Alice is randomly shown new information, she will not update her beliefs very much, since her beliefs are already very precise. If Bob is randomly shown new information, he will significantly revise his beliefs. However, since Bob doesn't base the price he sets on his beliefs about the market value, his list price won't change. The experimenter may then conclude that beliefs about the market value are not important for people's list price, even though we know that Alice sets her price exactly to her belief of the market value.

People whose actions depend on their beliefs will invest time and energy into forming precise beliefs, before the experiment takes place. Then, at the time of the experiment, the people who are most responsive to new information are the people for whom these beliefs are relatively unimportant to their decision-making. In this a way, a simple analysis of such an experiment will underestimate the extent to which beliefs about the

[6] Questions about how beliefs affect the choice to buy or sell a home have been studied by several information provision experiments, for example Armona et al. (2019) and Bottan and Perez-Truglia (2022a).

market value of a home affect the list price.

A Toy Model of Belief Formation via Costly Information Acquisition. These dynamics arise in a simple model where people make a choice about purchasing a costly signal that they use to update their beliefs.

Let Y be the action (e.g., list price of a house) and X denote beliefs (e.g., about the market value). People start with a prior belief distribution centered around π_i. The initial variance of their beliefs is $\sigma^2_{X_0}$ so that their beliefs are represented by the normal $\mathcal{N}(\pi_i, \sigma^2_{X_0})$. For simplicity, $\sigma^2_{X_0}$ is common. We will consider signals S drawn from a normal distribution $\mathcal{N}(\mu_S, \sigma^2_S)$. This is an assumption that people have the same information environment.

People are uncertain about their beliefs, and this uncertainty about their beliefs generates uncertainty about the action that they would like to take. People act to minimize the loss function $L_i(y, x) = D\big(y, Y_i(x)\big)$, for some distance function D, which is the disutility associated with taking action y when $X = x$. Intuitively, integrating $L_i(y, x)$ over the distribution of beliefs converts uncertainty about beliefs (i.e., what is the probability that $X = x$) into regret about actions (i.e., how far is the choice y from $Y_i(x)$, which is optimal when $X = x$). In this loss function, beliefs affect utility only through their effect on actions. There is no direct "psychic" cost of imprecise beliefs.

People choose $Y_i(x)$ following the rule $Y_i(x) = \tau_i x + U_i$, where τ_i and U_i vary across individuals, and have quadratic loss $D(a, b) = \big(a - b\big)^2$. They act to minimize their expected loss, which is simply the expectation of $L_i(y, x)$ with respect to X $\big($i.e. $\mathbb{E}_X\big[L_i(y, x)\big]\big)$.

When beliefs are given by the normal $\mathcal{N}(\overline{X}, \sigma^2_{X_0})$, the choice of Y that minimizes expected loss is simply $Y^* \equiv Y_i(\overline{X}) = \tau_i \overline{X} + U_i$. We can use this to further simplify the expression for expected loss and write

$$\mathbb{E}_X\big[L_i(Y^*, x)\big] = \mathbb{E}_X\big[D(Y_i(\overline{X}), Y_i(x))\big] \tag{6}$$

$$= \mathbb{E}_X\big[\big((\tau_i \overline{X} + U_i) - (\tau_i x + U_i)\big)^2\big] = \tau_i^2 \sigma^2_X \tag{7}$$

The disutility generated by uncertainty about X is increasing in both the variance of the belief distribution and the magnitude of the causal effect of beliefs on the outcome. This expression allows us to study the information acquisition problem.

I endogenize belief formation by allowing people to pay a fixed cost C to view a signal that is centered around the unknown true value. They then update beliefs following

the normal-normal Bayesian learning formula we have been working with throughout. When a person's beliefs are given by $\mathcal{N}(\overline{X}, \sigma_x^2)$, her loss is given recursively by

$$V_i(\overline{X}, \sigma_x^2) = \min \left\{ \mathbb{E}_X[L_i(Y_i(\overline{X}), x)], \mathbb{E}_S[V_i(X'(s)), \sigma_{X'}^2)] + C] \right\} \tag{8}$$

Where $\sigma_{X'}^2 = \frac{\sigma_X^2 \sigma_S^2}{\sigma_X^2 + \sigma_S^2}$ and the expectation $\mathbb{E}_S$ is the expectation with respect to the signal. Notice that in this model, the benefit of the signal comes from the fact that the posterior variance is less than the prior variance as long as the prior distribution is not already degenerate.

Solving this recursive problem gives the equilibrium condition

$$\begin{aligned} V_i(\overline{X}, \sigma_x^2) &= \min \left\{ \tau_i^2 \sigma_X^2, \tau_i^2 \sigma_{X'}^2 + C \right\} \\ \tau_i^2 \sigma_X^2 &= \tau_i^2 \sigma_{X'}^2 + C \end{aligned} \tag{9}$$

In equilibrium, agents will be indifferent between paying the fixed cost to obtain new information and living with the uncertainty they have.[7] Replacing $\sigma_{X'}^2$ with its definition, and recalling that $1 - \frac{\sigma_S^2}{\sigma_S^2 + \sigma_X^2} = \alpha_i$ we obtain the following equality

$$\alpha_i \tau_i^2 \sigma_X^2 = C \tag{10}$$

Agents for whom the outcome is very sensitive to the beliefs (τ_i^2 is very large) will update their information until $\sigma_X^2 \alpha_i$ is small.[8] On the other hand, agents for whom the outcome is not sensitive to beliefs (τ_i^2 is small) will stop after seeing fewer signals, so that $\sigma_X^2 \alpha_i$ is relatively large.

We can see in this toy model how the causal relationship of interest affects the formation of beliefs before the experiment takes place. People whose actions depend more on their beliefs will be more willing to pay to obtain new information, and will

[7] To ease exposition, I have ignored integer constraints that will, in general, prevent this from holding with equality. People will purchase signals until the next signal reduces their expected loss by less than the cost of the signal and will generally be strictly worse off if they buy another signal, not indifferent. This technicality makes exposition more cumbersome without any conceptual payoff.

[8] Notice that since $\alpha_i \equiv \frac{\sigma_X^2}{\sigma_S^2 + \sigma_X^2}$, α_i and σ_X^2 move together. That is, holding fixed σ_S^2, an increase in σ_X^2 implies an increase in α_i and vice-versa.

therefore have more precise beliefs. In a Bayesian updating model, people with more precise beliefs will be less responsive to new information. The amount of variation in beliefs that can be induced by experimentally providing new information is therefore not random. We will now consider what this means for standard empirical specifications and characterize TSLS weights as a function of the learning rate α_i.

3. What Parameters do Common Empirical Specifications Target?

This model of belief formation can help to guide the interpretation of common empirical specifications. We have discussed two common empirical settings: the *active* design where treatment is assignment to the high signal and the *passive* design where treatment is assignment to new information.

Sometimes, these studies measure reduced form intent-to-treat parameters. As has become more widely appreciated recently (for example by Giaccobasso et al. (2022)), these parameters can be difficult to interpret. These intent-to-treat (ITT) parameters are a combination of the effect of beliefs on behaviors and the first stage effect of the information provision on beliefs. The ITT can be small if beliefs have only a weak effect on behavior, or if the information provision has only a weak effect on beliefs. Even if the effect of beliefs on behaviors always has the same sign, the reduced form effect of information provision will have a different sign for people who overestimate and underestimate.

The reduced form effect of randomization into treatment is policy-relevant when the counterfactuals of interest are about future information provision. It is more difficult to interpret when the relationship of interest is the effect of beliefs on behavior. Since this paper focuses on identifying the causal effect of beliefs on behavior, I will focus on empirical specifications that (aim to) target parameters that are informative about the effect of beliefs on behavior.

The literature does this with various instrumental variables specifications. In the simplest specifications, random assignment is used as an instrument for beliefs. The only difference between these specifications in active and passive designs is the construction of the treatment indicator: in active designs it indicates assignment to the high signal, in passive designs it indicates assignment to receive a new signal. Alternative specifications use functions of the prior and/or the signal to construct instruments. In any case, the coefficient of interest is then the coefficient on beliefs in a two-stage least squares regression. We will consider each design in turn.

3.1. Common Empirical Specifications in Active Designs

Recall that in the active design, participants are randomly assigned to a high signal $S_i(2)$ or low signal $S_i(1)$. The potential outcome for the posterior is thus

$$X(z) = X^0 + \alpha_i\left(S_i(z) - X_i^0\right)$$

Using Treatment Assignment as an Instrument for Beliefs. The simplest specification uses the indicator for randomization into the high signal group $Z_i^H \equiv \mathbb{1}\left(Z_i = 2\right)$ as the instrument. The associated estimand is

$$\beta_{\text{active}}^{IV} \equiv \frac{\text{Cov}\left(Z_i^H, Y_i\right)}{\text{Cov}\left(Z_i^H, X_i\right)} \tag{11}$$

$$= \frac{\mathbb{E}\left[\tau_i \alpha_i\left(S_i(2) - S_i(1)\right)\right]}{\mathbb{E}\left[\alpha_i\left(S_i(2) - S_i(1)\right)\right]} \tag{12}$$

$$= \mathbb{E}\left[\tau_i \cdot \underbrace{\alpha_i\left(S_i(2) - S_i(1)\right)\Omega_a^{-1}}_{\text{weights}}\right] \tag{13}$$

Where $\Omega_a \equiv \mathbb{E}\left[\alpha_i(S_i(2) - S_i(1))\right]$ is a normalizing constant that ensures the weights integrate to one.

I will use this format to study empirical specifications throughout, so I will briefly comment here. The first equality in (11) defines the TSLS target parameter. The second expression in (12) is the Wald estimand, which expresses the TSLS target parameter in terms of potential outcomes. Finally, my preferred expression in (13) is simply a rearranged version of (12) that highlights the interpretation of this parameter as a weighted average of individual causal effects τ_i. Notice that the weights are exactly the strength of the (heterogeneous) individual first stage normalized by the average strength of the first stage; Ω_a is simply the denominator in (12).

If the treatment arms are coded so that $z = 2$ denotes a "high" signal and $z = 1$ denotes a "low" signal, so that $S_i(1) < S_i(2)$ for all i, then by construction $\alpha_i(S_i(2) - S_i(1)) > 0$, and $\beta_{\text{active}}^{IV}$ is a causal parameter.

Constructing an Instrument with the Difference in Signals. The estimator $\beta_{\text{active}}^{IV}$ is most common in settings where the signals are common, for example Settele (2022), which

randomizes over high and low signals of the gender wage gap. However, when the signals vary across people or groups as in Bottan and Perez-Truglia (2022b) and Roth et al. (2022), an alternative specification uses the constructed instrument $\Delta_S \equiv S_i(2) - S_i(1)$.

Two-stage least squares regression then targets the parameter

$$\beta^{IV}_{\Delta_S} \equiv \frac{\mathrm{Cov}\left(\Delta_S, Y_i\right)}{\mathrm{Cov}\left(\Delta_S, X_i\right)} \tag{14}$$

$$= \frac{\mathbb{E}\left[\tau_i \alpha_i \left(S_i(2) - S_i(1)\right)^2\right]}{\mathbb{E}\left[\alpha_i \left(S_i(2) - S_i(1)\right)^2\right]} \tag{15}$$

$$= \mathbb{E}\left[\tau_i \cdot \underbrace{\alpha_i \left(S_i(2) - S_i(1)\right)^2 \Omega^{-1}_{\Delta_S}}_{\text{weights}}\right] \tag{16}$$

Where $\Omega_{\Delta_S} \equiv \mathbb{E}\left[\alpha_i \left(S_i(2) - S_i(1)\right)^2\right]$. Notice that this has positive weights whenever $\alpha_i > 0$, and does not rely on ordering categorizing signals into a "high" and "low" signal such that $S_i(2) > S_i(1)$ for all i.

The simple model of belief formation presented in Section 2 predicts that people with the weakest causal effects of beliefs on behavior will also have the least precise beliefs, and will accordingly update their beliefs the most in response to new information. For this reason, these TSLS estimands will understate the strength of the effect of beliefs on behavior.

Additionally, note that when the potential signals $S_i(2)$ and $S_i(1)$ are person-specific, this estimand puts relatively more weight on people with large difference in their potential signals $(S_i(2) - S_i(1))$. If the variation in signals is assumed to be independent of the heterogeneity (as in Roth et al., 2022), then the weights further simplify to be proportional only to the learning rate α_i.

3.2. Common Empirical Specifications in Passive Designs

In the passive design, participants are randomly assigned to receive a signal S_i in the treatment group, or no new information in the control group. The potential outcome for the posterior is thus

$$X_i(z) = X_i^0 + \mathbb{1}\left(z = 1\right) \alpha_i \left(S_i - X_i^0\right) \tag{17}$$

Using Treatment Assignment as an Instrument for Beliefs. The simplest TSLS specification that uses the treatment indicator as an instrument for beliefs is given by

$$\beta^{IV}_{\text{passive}} \equiv \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)} \tag{18}$$

$$= \frac{\mathbb{E}\left[\tau_i \alpha_i (S_i - X_i^0)\right]}{\mathbb{E}\left[\alpha_i (S_i - X_i^0)\right]} \tag{19}$$

$$= \mathbb{E}\left[\tau_i \cdot \underbrace{\alpha_i (S_i - X_i^0) \Omega_p^{-1}}_{\text{weights}}\right] \tag{20}$$

Where $\Omega_p \equiv \mathbb{E}\left[\alpha_i (S_i - X_i^0)\right]$ is a normalizing constant. Notice that these weights are only (weakly) positive when $\alpha_i(S_i - X_i^0)$ has the same sign for all participants. This is the instrument monotonicity assumption. This estimand is only a convex combination of individual effects when the information treatment shifts beliefs in the same direction for all participants. In a Bayesian model of the first stage, this is equivalent to the requirement that the signal is either above (or below) the prior for every person, which is often hard to justify empirically. [9] For this reason, (unconditional) instrument monotonicity is often difficult to justify in this setting.

Using Priors to Ensure Positive Weights in Passive Designs. The current approach to ensure that the weights are (weakly) positive is to use information on prior beliefs. The simplest way that information on priors can be incorporated is by splitting the sample based on the direction that the signal shifts beliefs. Consider estimating the regression in (18) separately for the groups with priors above and below the signal.

$$\beta_+^{\text{split}} \equiv \frac{\text{Cov}(Z_i, Y_i \mid S_i - X_i^0 > 0)}{\text{Cov}(Z_i, X_i \mid S_i - X_i^0 > 0)} \tag{21}$$

$$= \frac{\mathbb{E}\left[\tau_i \alpha_i |S_i - X_i^0| \;\middle|\; S_i - X_i^0 > 0\right]}{\mathbb{E}\left[\alpha_i |S_i - X_i^0| \mid S_i - X_i^0 > 0\right]} \tag{22}$$

$$= \mathbb{E}\left[\tau_i \cdot \alpha_i |S_i - X_i^0| \Omega_+^{-1} \;\middle|\; S_i - X_i^0 > 0\right] \tag{23}$$

[9] For example Cullen et al. (2022), Cullen and Perez-Truglia (2022), and Fuster et al. (2022) observe people with prior beliefs above and below the signal.

and symmetrically

$$\beta_{-}^{\text{split}} \equiv \frac{\text{Cov}\left(Z_i, Y_i \mid S_i - X_i^0 < 0\right)}{\text{Cov}\left(Z_i, X_i \mid S_i - X_i^0 < 0\right)} \tag{24}$$

$$= \frac{\mathbb{E}\left[\tau_i \cdot \alpha_i \left|S_i - X_i^0\right| \;\middle|\; S_i - X_i^0 < 0\right]}{\mathbb{E}\left[\alpha_i \left|S_i - X_i^0\right| \mid S_i - X_i^0 < 0\right]} \tag{25}$$

$$= \mathbb{E}\left[\tau_i \cdot \alpha_i \left|S_i - X_i^0\right| \Omega_{-}^{-1} \;\middle|\; S_i - X_i^0 < 0\right] \tag{26}$$

Where $\Omega_{+} \equiv \mathbb{E}\left[\alpha_i \left|S_i - X_i^0\right| \mid S_i - X_i^0 > 0\right]$ and $\Omega_{-} \equiv \mathbb{E}\left[\alpha_i \left|S_i - X_i^0\right| \mid S_i - X_i^0 < 0\right]$ are constants. Both β_{+}^{split} and β_{+}^{split} are causal parameters since they can be interpreted as non-negatively weighted averages of individual partial effects τ_i. One natural choice of weights to aggregate β_{+}^{split} and β_{+}^{split} into a single parameter is the size of each group. Conditional on the sign of the difference between the prior and the signal, all beliefs shift in the same direction (towards the signal) and the negative weights are eliminated.

Notice that the priors X_i^0 are used only to construct the subsamples. After conditioning on the sign of $S_i - X_i^0$, the same TSLS regression as in (18) is estimated. These estimands have non-negative weights under the assumption that people update beliefs weakly towards the signal. This is implied by the Bayesian updating model we maintain throughout, but such an assumption can also be made directly. If this assumption is slightly strengthened so that people update beliefs weakly towards, but not past, the signal, then feasible estimators for the split estimators (21), (24) can be constructed even when the prior belief is not directly observed. If the posterior lies between the prior and the signal, the direction that the signal shifted beliefs can be inferred from the posterior and the signal. See Appendix A for further discussion of this strategy to ensure positive weights in a passive design, even when priors are not elicited and monotonicity does not hold.

Another way that information on the prior can be incorporated is by constructing an exposure-weighted instrument (Armona et al., 2019; Cullen and Perez-Truglia, 2022). The exposure-weighted instrument

$$\Delta_Z \equiv \mathbb{1}\left(Z = 1\right)\left(S_i - X_i^0\right) \tag{27}$$

interacts the treatment indicator with the difference between the prior and the signal. In order to be uncorrelated with the structural error term U_i, this instrument needs to be re-centered; this can be done manually or by including $(S_i - X_i^0)$ as a control

as in Cullen and Perez-Truglia (2022). After re-centering, the instrument is

$$\widetilde{\Delta}_Z \equiv \left(\mathbb{1}\left(Z=1\right) - \mathbb{P}\left[Z=1\right]\right)(S_i - X_i^0) \tag{28}$$

and

$$\beta_\Delta^{IV} \equiv \frac{\mathrm{Cov}\left(\widetilde{\Delta}_Z Y_i\right)}{\mathrm{Cov}\left(\widetilde{\Delta}_Z X_i\right)} \tag{29}$$

$$= \frac{\mathbb{E}\left[\tau\alpha_i(S_i - X_i^0)^2\right]}{\mathbb{E}\left[\alpha_i(S_i - X_i^0)^2\right]} \tag{30}$$

$$= \mathbb{E}\left[\tau \cdot \alpha_i(S_i - X_i^0)^2 \Omega_\Delta^{-1}\right] \tag{31}$$

Where $\Omega_\Delta \equiv \mathbb{E}\left[\alpha_i(S_i - X_i^0)^2\right]$ is a normalizing constant. Instead of weighting individuals proportional to the absolute value of the first stage (as in the split sample approach), the exposure-weighted instrument weights them proportional to $\alpha_i(S_i - X_i^0)^2$, which is quadratic in the difference between the signal and the prior. These weights are non-negative, but put even more weight on people whose priors are further from the signal.

3.3. Interpreting Parameters in Common Empirical Specifications

When researchers are interested in making more general inferences about the effect that beliefs have on the behavior of interest, these parameters can be difficult to interpret, even when they are non-negatively weighted averages.

All of these parameters aggregate individual partial effects τ_i by weighting them proportionally to α_i, putting the most weight on those with the least precise priors (large α_i). In active designs, it also puts more weight on people with a large difference in their high and low signals (large $S_i(2) - S_i(1)$). In passive designs, the standard parameters also put more weight on those with the least accurate beliefs (large $|S_i - X_i^0|$). Some TSLS specifications ensure that weights are non-negative by interacting the treatment indicator with a measure of "exposure" to the signal, like the difference between signals Δ_S or the difference between the signal and the prior Δ_Z. These introduce a term proportional to this difference squared, which is non-negative, but puts greater weight on outliers.

In the toy model discussed in Section 2, beliefs are formed endogenously through a

process of costly information acquisition. People have inaccurate and imprecise beliefs precisely because they have small individual partial effects (small τ_i^2); when beliefs are an important determinant of the behaviors (large τ_i^2), people expend more effort forming accurate and precise beliefs. In this environment, parameters with weights proportional to the strength of the shift in beliefs will be attenuated and underestimate the magnitude of the population average effect.[10]

Alternative models of the relationship between belief formation and the effects of beliefs on behaviors can be used to relate casual parameters estimated via TSLS to the APE. For example, Fuster et al. (2022) allow variation in the learning rate to come from a more complicated model that adds dynamics of rational inattention to costly information acquisition. Any model that makes predictions about the covariance between the learning rate α_i and the causal effect of beliefs on behavior τ_i can be used instead of the model in Section 2. Only in the special case when the heterogeneity in the first stage is uncorrelated with individual causal effects of interest will TSLS identify the average partial effect.

4. Identifying the Average Partial Effect of Beliefs on Behavior

Since the parameters estimated in common TSLS specifications can understate the effect of beliefs on behaviors, researchers hoping to learn more generally about the effect of beliefs on behaviors may be interested in targeting alternative parameters that are informative about the general population, and not just the subset whose beliefs are responsive to new information. One contribution of this paper is to show that the such a parameter–the average partial effect $\mathbb{E}[\tau_i]$–is point identified in active designs that elicit priors.

Intuitively, conditional on the prior and the assigned signal, the remaining variation in the posterior depends only on α_i. In contrast, the APE is not identified in the passive design. The learning rate in the passive control group is not identified, since the researcher never observes how they update in response to new information.

[10]When the effects of beliefs on behaviors τ_i always have the same sign, then this attenuation effect biases the estimate towards zero. If the τ_i are of either sign, then this bias will still put more weight on people with smaller individual effects (in magnitude), but it is not possible to sign the average partial effect. In general, parameters with an interpretation as a non-negatively weighted average are only informative about the sign of the average effect when all individual effects have the same sign.

4.1. Using a Control Function to Identify the Average Partial Effect in Active Designs

Recall that in the active design, participants are randomly assigned to a high signal $S_i(2)$ or low signal $S_i(1)$. The potential outcome for the posterior is thus

$$X_i(z) = X_i^0 + \alpha_i\left(S_i(z) - X_i^0\right)$$

We will now be able to identify the APE by observing the prior, the posterior, and the behavior of interest. We will use a control function to identify the average partial effect $\mathbb{E}\left[\tau_i\right]$. This applies results in Masten and Torgovitsky (2016) to a case with additional control variables that can enter the outcome equation non-linearly and are not assumed to be exogenous, in the sense that the controls $W_i \equiv \begin{bmatrix} X_i^0 & S_i(1) & S_i(2) \end{bmatrix}$ are not assumed to be independent of U_i.

We will condition on the prior, the potential signals and the assigned treatment and then use the remaining variation in X to identify the conditional rank of α_i. Following Masten and Torgovitsky (2016), we will use a two-step approach to identify $\mathbb{E}\left[\tau_i\right]$. First, construct a control function R_i such that $\frac{\mathrm{Cov}\left(Y_i, X_i | R_i = r\right)}{\mathrm{Var}\left(X_i | R_i = r\right)} = \mathbb{E}\left[\tau_i \mid R_i = r\right]$. Then, integrate over the support of the control function to identify $\mathbb{E}\left[\tau_i\right]$. We can construct such a control function as follows:

$$R_{i1} = F_{\alpha|W,Z}(\alpha_i) = \begin{cases} F_{X|W,Z}(X_i) & S_i(Z_i) - X_i^0 > 0 \\ 1 - F_{X|W,Z}(X_i) & S_i(Z_i) - X_i^0 < 0 \end{cases} \tag{32}$$

$$R_i = \begin{bmatrix} R_{i1} & W_i \end{bmatrix} = \begin{bmatrix} F_{\alpha|W}(\alpha_i) & W_i \end{bmatrix} \tag{33}$$

Conditional on the prior and potential signals W_i, and the assigned treatment Z_i, all the remaining variation in the posterior X_i is generated by the learning rate α_i. Further, it is clear from (2) that the unobservable α_i is not identified among a control group that is not shown new information.

The formal identification statement is that the control function R_i is such that $\frac{\mathrm{Cov}\left(Y_i, X_i | R_i = r\right)}{\mathrm{Var}\left(X_i | R_i = r\right)} = \mathbb{E}\left[\tau_i \mid R_i = r\right]$. If $\mathrm{Var}\left(X_i \mid R_i = r\right) \neq 0$ for all r, then it follows immediately from the law of iterated expectations that $\mathbb{E}\left[\mathbb{E}\left[\tau_i \mid R\right]\right] = \mathbb{E}\left[\tau_i\right]$ so $\mathbb{E}\left[\tau_i\right]$ is identified. If $\alpha_i > 0$ for all i, which in the Bayesian model is equivalent to assuming that prior beliefs are never degenerate, then it is sufficient to design the experiment

such that $S_i(1) \neq X_i^0$, $S_i(2) \neq X_i^0$, $S_i(1) \neq S_i(2)$, and $0 < \mathbb{P}[Z_i = 1] < 1$ to ensure that $\mathrm{Var}(X_i \mid R_i = r) \neq 0$. If any of these do not hold, then the conditional average partial effect can be identified for the subset $r \in \mathcal{R}$ such that $\mathrm{Var}(X_i \mid R_i = r) \neq 0$. The two-stage control function is an alternative way to use the variation induced by the instrument; instrument relevance is thus still required.

Intuitively, conditioning on the prior and the potential signals is like isolating a sub-experiment that is restricted to people with the same prior and the same potential signals. In this sub-experiment, the variation in the posterior X_i among people in the same treatment arm is due only to differences in the learning rate, and so the rank of the learning rate can be inferred from this variation. Still conditioning on the prior and potential signals W_i, when we condition now on the rank of the unobservable R_{i1}, the only variation in the posterior comes from treatment assignment. I can now state the identification result formally.

Assumption 1.

(a) ***Instrument Exogeneity*** $Z_i \perp\!\!\!\perp (X_i^0, S_i, U_i, \alpha_i, \tau_i)$

(b) ***Linear Outcome (eq. 1)*** *The outcome is linear in beliefs; partial effects are heterogeneous.* $Y_i = \tau_i X_i + U_i$

(c) ***Linear Updating (eq. 2)*** *Beliefs are updated following* $X_i(z) = X_i^0 + \alpha_i(S_i(z) - X_i^0)$

(d) ***Relevance*** *The instrument always induces variation in beliefs.*
$\alpha_i \in (0,1)$; $S_i(1) \neq X_i^0$, $S_i(2) \neq X_i^0$, $S_i(1) \neq S_i(2)$ *and* $\mathbb{P}[Z_i = 1] \in (0,1)$

(e) ***Existence*** $\mathbb{E}[\tau_i]$ *and* $\mathrm{Var}(X_i)$ *exist and are finite*

Proposition 1. *Under Assumption 1,* $\frac{\mathrm{Cov}(Y_i, X_i | R=r)}{\mathrm{Var}(X_i | R_i = r)} = \mathbb{E}[\tau_i \mid R_i = r]$.

The proof is an immediate application of the argument in Masten and Torgovitsky (2014) conditional on the sign of the difference between the signal and the prior (see Appendix B).

We can then identify $\mathbb{E}[\tau_i]$ by integrating over the support of R. Since W_i is directly observed, and (32) shows that the remaining entry R_{i1} is also identified, the regression of Y_i on X_i conditional on $R_i = r$ is feasible. Of course, since $\mathbb{E}[\tau_i | R_i = r]$ is identified, heterogeneity analysis is immediately available by aggregating over only a subset of possible values of the control function. For example, applied researchers also seem to be interested in average effects for different groups based on observables.

For ease of exposition, I have maintained in Assumption 1.c that belief updating is linear, though it is not necessarily Bayesian since there is no structure on the learning rates α_i.[11] However, it is not necessary for belief updating to be linear. The APE will still be identified as long as beliefs can be expressed by a function $X_i = h(X_i^0, S_i, Z_i, \alpha_i)$ that is strictly increasing in α_i when the signal is above the prior and strictly decreasing when the signal is below the prior.[12] This structure is equivalent to the statement in (32) that identifies the conditional rank of α_i through the conditional rank of the posterior X_i. For example, a model where people discount more surprising signals so that $X_i = X_i^0 + \alpha_i \frac{(S_i(Z_i) - X_i^0)}{D(S_i(Z_i), X_i^0)}$ for some distance function D will satisfy this condition. Notice that this structure need not imply even conditional monotonicity. The APE will still be identified in a model that allows α_i to take negative values so that people update their beliefs away from the signal.

When the outcome equation is nonlinear, the control function will still ensure that there is no bias due to correlation between the strength of the first stage (i.e., α_i) and the second stage (τ_i). Instead of the average partial effect, the control function estimator will recover a mean-squared error optimal linear approximation of the average partial effect (Masten and Torgovitsky, 2016).

The Average Partial Effect is Not Identified in Passive Designs. In active designs, the rank of the learning rate is identified from the variation in the posterior after conditioning on the prior, the potential signals, and treatment assignment. This step is not possible in a passive design; in the passive control group, $X_i = X_i^0$ and thus the learning rate is not identified. Thus, the key first element of the control function is not identified as in (32).

The average partial effect is a person-weighted average of individual effects. When such equally-weighted averages cannot be identified, researchers often will instead aim to estimate a (weakly) positively weighted average of individual effects (i.e., a causal parameter). In Appendix A, I provide a novel argument to identify a causal parameter in passive designs that does not require eliciting priors or monotonicity.

While the average partial effect is not generally identified in passive designs, a model like the one in Section 2 that relates the causal effects of interest to the learning rate can be used to sign the bias of TSLS estimates relative to the unweighted average.

[11] Thus any model of the variation in α_i can be used. In Section 5, I will show how this approach allows for models of the variation in α_i to be empirically tested.

[12] Conditional on the sign of the difference between the signal S_i and the prior X_i^0, h is monotonic in the unobservable α_i and the results in Masten and Torgovitsky (2014) can be used.

5. An Application to the Effect of Beliefs on Demand for Public Policy

Settele (2022) studies the effect of beliefs about the gender wage gap on policies aimed at increasing gender equality. Settele randomly assigns some participants to receive a high signal of women's relative wages – that women's wages are 94% of men's wages – and some to a low signal of women's wages – that they are 74% of men's wages. [13] The main estimates use the variation in the signal to estimate the causal effects of beliefs. There is also a pure control group that is not shown any new information. Since the learning rate is not identified for the pure control group that receives no information, I will focus on the active design: that is, the comparison between the groups that receive high and low signals of relative earnings for women.

In my preferred specification, the APE of a standard deviation shift in beliefs on a policy demand index is roughly 40% larger in magnitude than the corresponding TSLS estimate. The imprecisely estimated heterogeneous treatment effects are also consistent with the theoretical predictions. The average partial effect for people in the top fifth of the learning rate α_i distribution is close to zero (0.01); and the average partial effect for the bottom 80% of the learning rate (-0.19) is roughly double the corresponding 2SLS estimate (-0.105).

The TSLS estimand puts disproportionate weight on people with the largest learning rates (see Section 3). Consistent with the theoretical predictions, these people adjust their beliefs the most in response to new information precisely because these beliefs are not important for their choices. In this example, the 2SLS estimand places roughly equal weight on the top 20% and the bottom 80% and understates the average partial effect by roughly 1/3rd.

In the rest of this section, I present the results from the control function and TSLS and show that they are consistent with the predictions of the model in Section 2.

5.1. Estimating the Control Function

In the first stage, I will use a series of quantile regressions to approximately estimate the conditional ranks of the posterior X_i given the prior and potential signals W_i and treatment assignment Z_i. Since the potential signals are common, this simplifies to simply conditioning on the prior and the assigned signal. See Appendix C.1 for a detailed

[13] These estimates come from the most recent CPS and ACS survey at the time of the experiment, and so are generated without deception. Akesson et al. (2022) and Bottan and Perez-Truglia (2022b) also use this source randomization strategy to generate variation in potential signals without deception. See Settele (2022) for more information about the design of the experiment.

discussion of how the first stage estimates of the conditional ranks R_{i1} of the learning rate α_i are estimated. In Appendix C.2, I show that results are qualitatively similar when using a non-parametric approach to estimate the first stage, though estimation is noisier.

Then, in the second stage, I condition on the first-stage estimates of the conditional rank of the learning rate using a kernel. Then, I estimate the intermediate conditional average partial effect $\mathbb{E}\left[\tau_i \mid F_{\alpha_i|W_i}(\alpha_i) = r_1\right]$ by regressing the policy index Y_i on the posterior belief X_i and the prior X_i^0.[14] Conditional on the learning rate and the prior, the only remaining variation in beliefs comes from the value of the randomly assigned signal. Then, I estimate the average partial effect by numerically integrating over the distribution of ranks. To estimate standard errors, I bootstrap the two-stage procedure following Masten and Torgovitsky (2016).

Additional Controls from Settele (2022) Are Omitted. To avoid complications from including additional covariates, I ignore the additional controls that Settele (2022) includes in the main specification reported in Table 5 of the paper. In Appendix C.4, I compare results in the specification without controls that I will consider to the original paper specification. Point estimates are somewhat larger in magnitude without these additional controls, but results are qualitatively similar. Thus, comparisons between the APE that I estimate and TSLS estimates will use results from this more parsimonious specification, and not the results reported in Settele (2022).

5.2. Empirical Evidence that TSLS Underestimates the APE

The main TSLS specification in Settele (2022) Table 5 uses an indicator for assignment to the "low" signal as an instrument for posterior beliefs. Since the signals are common ($S_i(z) = S(z)$), TSLS targets

$$\mathbb{E}\left[\tau_i \cdot \widetilde{\alpha}_i\right] \tag{34}$$

where $\widetilde{\alpha}_i \equiv \frac{\alpha_i}{\mathbb{E}\left[\alpha_i\right]}$ is the learning rate α_i normalized by the mean $\mathbb{E}\left[\alpha_i\right]$ so that

[14] If the linear updating model is correctly specified, then it is sufficient to control for the prior linearly, which eases estimation. It is possible to condition on the prior non-parametrically by using, for example, a two-dimensional kernel to condition on both $F(\alpha_i)$ and the prior and then regressing the outcome directly on the prior in the intermediate regression. This is a more data intensive approach that will use fewer observations for each intermediate regression. In Appendix C.3, I show that including a quadratic term in the prior to this regression changes the estimate of the APE only at the third decimal. This suggests that using a linear control for the prior at this step in estimation is not overly restrictive.

these weights integrate to 1. [15] Under the Bayesian learning assumption maintained throughout (in particular the assumption that $0 \leq \alpha_i$), this is a non-negatively weighted average. Since these weights are given by the learning rate α_i, the toy model in Section 2 predicts that the TSLS estimates will be smaller in magnitude than the average partial effect. Replicating the main specification from the paper without additional controls yields an estimate of –0.105.[16]

In Figure 1, I report an estimate of the average partial effect of beliefs on the policy demand index. The control function estimate of the APE is –0.146, which is roughly 40% larger in magnitude than the TSLS estimate of –0.105. I also plot estimates of the conditional average partial effect for different ranks of the learning rate α_i. The conditional average partial effect is the average partial effect for people at the r_1th rank of the learning rate $\mathbb{E}\left[\tau_i \mid F_{\alpha_i|W_i}(\alpha_i) = r_1\right]$. The estimates of the conditional average partial effect are noisy, but the estimates are consistent with the predictions of the model in Section 2. In particular, notice that the conditional average partial effect is close to zero for the top 20% of the learning rate distribution.

Conditional average partial effects are estimated at four hundred values equally spaced over the interval $(0, 1)$. To ease interpretation, I present two binned scatter plots after collapsing estimates of these heterogeneous effects into three (Panel A) or five (Panel B) equally-sized bins. For transparency, I present every intermediate estimate in panel C. Using fewer bins smooths out the noise in the underlying estimates of the conditional average partial effect at the cost of potentially suppressing interesting variation.

The gray regions at either end of the figure indicate the intervals within which the kernel is truncated. Since these estimates are particularly noisy, I exclude them from the bins in Panels A and B. The average partial effect is simply the average of the 400 estimates reported in Panel C, including the noisier estimates in the gray regions.

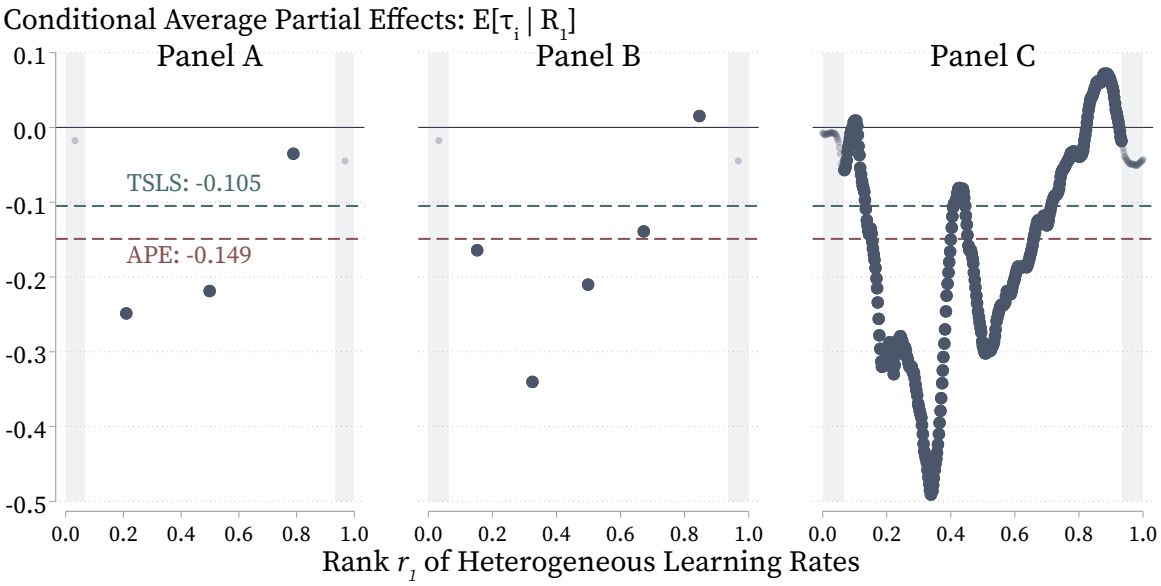
As the theory predicts, the people with the smallest effects of beliefs on outcomes have the largest learning rates and also receive the most weight in TSLS. The difference between the average effect in the top third (-0.031) and the bottom two-thirds (-0.210) of the learning rate distribution is large (-0.179) and robustly estimated.[17] This is consistent with the model developed in Section 2; people with the weakest effects of beliefs on

[15] Since the difference in signals is a constant, so they cancel from both the individual weights and the normalization term in the general expression in (13).

[16] Following the original paper, estimates throughout are scaled to be relative to a standard deviation change in beliefs.

[17] The 5th percentile of the bootstrapped distribution of this difference is -0.095.

FIGURE 1. 2SLS Estimates Understate the Average Effect of Beliefs



Notes: This figure plots estimates of conditional average partial effect $\mathbb{E}\left[\tau_i \mid F_{\alpha_i|W_i}(\alpha_i) = r_1\right]$ for different ranks r_1 of the learning rate α_i. The TSLS estimate is smaller in magnitude than the average partial effect because TSLS weights observations proportionally to their learning rate α_i, and people with the weakest effects of beliefs have the largest learning rates. Panel A and Panel B present results after collapsing to three and five equally-sized bins. Panel C displays all 400 underlying estimates. The shaded region denotes where the kernel begins to truncate due to the boundaries at 0 and 1. These regions are excluded when constructing the binned averages, instead the average estimates within these regions are plotted in light gray.

outcomes do not form precise beliefs and therefore update the most in response to new information. At the most aggressive level of smoothing, the conditional average partial effect appears to be roughly upward sloping in the rank of the learning rate (Panel A).

At a more intermediate level of smoothing, the conditional average partial effect appears to have a U-shape (Panel B). This may be because the estimates are particularly noisy for the people with the lowest ranks of the learning rate. However, it may also be that a richer model of belief formation is necessary to explain the behavior of people with the lowest ranks of the learning rate. Perhaps a model that also includes rational inattention in the spirit of Fuster et al. (2022) is needed to explain the behavior of people with the lowest learning rates. In such a model, some people with weak effects of beliefs on behavior would not update their beliefs at all in response to new information

because of costs of processing information. This dynamic may better explains behavior of people in the lowest part of the learning rate distribution. This would rationalize a U-shaped relationship between the rank of the learning rate and the effect of beliefs on behavior, but would complicate the expression for the learning rate.

This discussion highlights the power of this control function approach to estimate conditional average partial effects in addition to the overall average partial effect. Regardless of the model that generates heterogeneous learning rates, so long as the unobservable heterogeneity in the first stage can be captured in the learning rate α_i, the full conditional average partial effect curve can be estimated. This allows for a rich empirical investigation of the relationship between belief formation and the effect of beliefs on behavior.

6. Conclusion

This paper uses a workhorse Bayesian updating framework to study the identification of causal effects of beliefs on behavior in information provision experiments. If people are Bayesian, then the precision of their prior beliefs controls how much their beliefs shift when they are shown new information (i.e., the strength of the first stage).

Two-stage least squares estimates of the average effect of beliefs on behavior are biased by any covariance between the precision of prior beliefs and the causal effect of interests. Models of belief formation that relate the precision of beliefs to these causal effects can thus be used to sign the bias of IV estimates.

I show that a simple model with costly information acquisition predicts that the two-stage least squares estimates common in the literature will generally be smaller in magnitude than the unweighted average effect of beliefs. Common TSLS estimates understate the average strength of the causal effect of beliefs on behavior by giving more weight to people with weaker causal effects. This may explain puzzling results in the literature that find small or insignificant effects of beliefs on behavior despite having treatments that have a large effect on beliefs (Alesina et al., 2023; Kuziemko et al., 2015).

In an active experimental design that elicits priors, the Bayesian updating assumption on the first stage is sufficient to identify a linear approximation to the average partial effect. When the outcome equation is linear in beliefs, the average partial effect is identified without approximation error and has a structural interpretation as the average of the heterogeneous coefficients on beliefs.

I confirm these predictions in an application to a recent study of the effect of beliefs about the gender wage gap on demand for public policy (Settele, 2022). Using this control function estimator, I estimate that the average partial effect is more than 40% larger than the corresponding TSLS estimate. Consistent with the theoretical mechanism I propose, the difference between the APE and TSLS estimates appears to be driven by small causal effects among the population that updates their beliefs the most in response to new information. The average partial effect for the bottom two-thirds of the learning rate distribution is twice as large as the TSLS estimate. The APE for the third that updates their beliefs the most is indistinguishable from zero.

In passive designs, the Bayesian updating assumption enables a novel identification argument that does not rely on either collecting additional data or making a stronger monotonicity assumption in the first stage. Since the posterior lies between the prior and the signal, Bayesian updating allows us to infer the direction that the signal shifted beliefs from the posterior and the signal, without eliciting the prior.

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Appendix

A. Ensuring Positive Weights in Passive Designs With Unknown Priors

The dominant approach in the literature (see Section 3) to ensure that the weights are (weakly) positive is to incorporate information on the prior belief, either by splitting the sample (Cruces et al., 2013) or constructing an exposure-weighted instrument (Cullen and Perez-Truglia, 2022).

In this appendix, I provide a novel argument to identify a causal parameter that does not require eliciting priors or monotonicity. Instead, we use the assumption that people update their beliefs in the direction of the signal, but not past the signal. This follows from Bayesian updating, but Bayesian updating is not strictly necessary. Directly assuming that the posterior belief is a convex combination of the prior and the signal is sufficient for what follows.

This assumption identifies the same causal parameters that are targeted by β_{+}^{split} (23) and β_{-}^{split} (26). This is a weaker assumption than instrument monotonicity, which requires that the signal shifts beliefs in the same direction for all participants. This is a slightly stronger assumption than what was needed in Section 3 when priors were available. There we assumed that people update (weakly) towards the signal, but did not need to assume that they do not cross it.

Since the control group that is not shown a signal, we directly observe their prior: recall that $X_i(0) = X_i^0$. Since the signal is known, we can directly condition on the sign of $(S_i - X_i^0)$. The prior for the treated group is unknown and we observe only $X_i(1)$. But since we can rewrite the potential outcome equation in 2 as

$$S_i - X_i(1) = (1 - \alpha_i)(S_i - X_i^0)$$

and since $\alpha \in \big(0, 1\big)$ then

$$S_i - X_i(1) > 0 \iff (S_i - X_i^0) > 0$$

We used the Bayesian updating structure, but note this could be relaxed to include any model of updating such that the posterior lies between the prior and the signal.

Thus, although the regressions in (21) and (24) are not feasible, the following regressions are feasible and equivalent.

$$\beta_{+}^{\text{split}} = \widetilde{\beta}_{+}^{\text{split}} \equiv \frac{\text{Cov}\big(Z, Y \mid S_i - X_i > 0\big)}{\text{Cov}\big(Z_i, X_i \mid S_i - X_i > 0\big)} \tag{35}$$

$$\beta_{-}^{\text{split}} = \widetilde{\beta}_{-}^{\text{split}} \equiv \frac{\text{Cov}\big(Z, Y \mid S_i - X_i < 0\big)}{\text{Cov}\big(Z_i, X_i \mid S_i - X_i < 0\big)} \tag{36}$$

B. Proofs

B.1. The Average Partial Effect is Identified in Active Designs with Elicited Priors

This is a proof of proposition 1 that uses the structure of the specific information provision setting for the sake of exposition. The linear updating and Bayesian learning structure maintained throughout this paper is stronger than is necessary. Interested readers should see Masten and Torgovitsky (2016) for a more general proof and for a formal discussion of the more general conditions under which the APE is identified.

PROOF. First, use assumptions 1.b and 1.c to write

$$\begin{aligned}
&\mathrm{Cov}\big(Y_i, X_i \mid R_i = r\big) \\
&= \mathrm{Cov}\big(\tau_i\big(X_i^0 + \alpha_i(S_i(Z_i) - X_i^0)\big) + U_i, \big(X_i^0 + \alpha_i(S_i(Z_i) - X_i^0)\big) \mid R_i = r\big)
\end{aligned}$$

By definition $r \equiv \begin{bmatrix} r_1 & x^0 & s_1 & s_2 \end{bmatrix}$. Let $\widetilde{\alpha}(r)$ be such that $r_1 = F_{\alpha|w,z}(\widetilde{\alpha})$.[18] Substituting this in gives

$$\begin{aligned}
&= \mathrm{Cov}\big(\tau_i\big(x^0 + \widetilde{\alpha}(S_i(Z_i) - x^0)\big) + U_i, \big(x^0 + \widetilde{\alpha}(S_i(Z_i) - x^0)\big) \mid R_i = r\big) \\
&= \widetilde{\alpha}(r) x^0 \,\mathrm{Cov}\big(\tau_i, S_i(Z_i) \mid R_i = r\big) + \widetilde{\alpha}(r)^2 \,\mathrm{Cov}\big(\tau_i S_i(Z_i), S_i(Z_i) \mid R_i = r\big)
\end{aligned}$$

Notice that the remaining random variables are τ_i and the remaining variation in $S_i \mid R_i = r$. Writing S_i in terms of the switching equation makes it clear that all remaining variation in S_i comes from the treatment assignment Z_i. Recall that by definition $S_i = S_i(1)\mathbb{1}\big(Z_i = 1\big) + S_i(2)\mathbb{1}\big(Z_i = 2\big)$ so that $(S_i \mid R_i = r) = s_1\mathbb{1}\big(Z_i = 1\big) + s_2\mathbb{1}\big(Z_i = 2\big)$. From random assignment (1.a), $\mathrm{Cov}\big(\tau_i, S_i(Z_i) \mid R_i = r\big) = \mathrm{Cov}\big(\tau_i, s_1\mathbb{1}\big(Z_i = 1\big) + s_2\mathbb{1}\big(Z_i = 2\big) \mid R = r\big) = 0$. Substituting in the switching equation to the final term gives

$$= \widetilde{\alpha}(r)^2 \,\mathrm{Cov}\big(\tau_i\big(s_1\mathbb{1}\big(Z_i = 1\big) + s_2\mathbb{1}\big(Z_i = 2\big)\big), \big(s_1\mathbb{1}\big(Z_i = 1\big) + s_2\mathbb{1}\big(Z_i = 2\big)\big) \mid R_i = r\big)$$

so from random assignment

$$= \widetilde{\alpha}^2 \,\mathrm{Var}\big(\big(s_1\mathbb{1}\big(Z_i = 1\big) + s_2\mathbb{1}\big(Z_i = 2\big)\big) \mid R_i = r\big)\, \mathbb{E}\big[\tau_i \mid R_i = r\big]$$

Which finally gives

$$\begin{aligned}\operatorname{Cov}\big(Y_i, X_i \mid R_i = r\big) &= \widetilde{\alpha}_i^2 \operatorname{Var}\big(S_i(Z) \mid R = r\big)\, \mathbb{E}\big[\tau_i \mid R = r\big] \\ \operatorname{Var}\big(X \mid R = r\big) &= \operatorname{Var}\big(X_i^0 + \widetilde{\alpha}(S_i(Z) - X_i^0) \mid R = r\big) = \widetilde{\alpha}_i^2 \operatorname{Var}\big(S_i(Z) \mid R = r\big)\end{aligned}$$

Combining these results, we have that

$$\frac{\operatorname{Cov}\big(Y_i, X_i \mid R_i = r\big)}{\operatorname{Var}\big(X_i \mid R_i = r\big)} = \mathbb{E}\big[\tau_i \mid R_i = r\big]$$

where assumption 1.d ensures that

$$\operatorname{Var}\big(X_i \mid R_i = r\big) = \operatorname{Var}\Big(\big(x^0 + \widetilde{\alpha}(r)(S_i(Z_i) - x^0)\big) \mid R_i = r\Big) > 0$$

□

[18] So that $\widetilde{\alpha}(r)$ is defined without needing to restrict $F(\cdot)$, use $\widetilde{\alpha}(r) = \inf\big\{a \mid F_{\alpha|W}(a) = r\big\}$.

C. Additional Empirical Results and Alternative Specifications

In this section, I will provide some addition information about estimation of the control function in Section 5 and some results from alternative specifications.

C.1. Estimating the Rank of the Learning Rate via Quantile Regression

The potential outcomes for beliefs are given by (2), so that $X_i(z) = X_i^0 + \alpha_i\big(S_i(z) - X_i^0\big)$. Since signals are common, we have $S_i(z) = S(z)$ for all i. Thus, conditional on X_i^0 and Z_i, all the variation in $X_i(z)$ is due to variation in α_i. Notice, we can rearrange this expression to write

$$X_i - X_i^0 = \alpha_i\big(S(Z_i) - X_i^0\big) \tag{37}$$

where the only unknown in this expression is α_i. Notice then the conditional quantiles of $X_i - X_i^0$ given $S(Z_i) - X_i^0$ are exactly the conditional quantiles of α_i when $S(Z_i) - X_i^0 > 0$ and is one minus the conditional quantile when $S(Z_i) - X_i^0 < 0$. That is, when $S(Z_i) - X_i^0 > 0$ and the signal shifts beliefs in the positive direction, the largest values of $X_i - X_i^0$ correspond to the highest learning rates. However, when $S(Z_i) - X_i^0 < 0$ and the signal shifts beliefs in the negative direction, $X_i - X_i^0$ will also be negative, so the lowest values of $X_i - X_i^0$ will correspond to the highest learning rates.

To see this, let the qth quantile of Y be given by,

$$Q_Y(q) = F_Y^{-1}(q) = \inf\big\{\, y \mid F_Y(y) \geq q \big\} \tag{38}$$

Where F_Y is the CDF of Y. Then, let $Q_Y(q \mid X)$ be the qth conditional quantile of Y given X. Assume that $S(Z_i) - X_i^0 > 0$ and then

$$Q_{X_i - X_i^0}(q \mid S(Z_i) - X_i^0) = Q_{\alpha_i(S(Z_i) - X_i^0)}(q \mid S(Z_i) - X_i^0) = \alpha(q)\big(S(Z_i) - X_i^0\big) \tag{39}$$

where $\alpha(q)$ is the qth conditional quantile of α_i given $S(Z_i) - X_i^0)$. When $S(Z_i) - X_i^0 < 0$ results are analogous, except that $\alpha(q)$ is the (1-q)th conditional quantile. Since the posterior X_i, the prior X_i^0 and the signal $S(Z_i)$ are all observed, we can estimate the conditional quantiles of α_i via the quantile regression based rearrangement method of Chernozhukov et al. (2010). In my preferred specification, I estimate 250 quantile regressions of $X_i - X_i^0$ on $S(Z_i) - X_i^0$ at the quantiles $\frac{1}{251} \ldots \frac{250}{251}$. Then, for each observation, I calculate the share of these fitted values that are below the observed value of $X_i - X_i^0$.

This share is an estimate of the cumulative distribution evaluated at α_i. [19]

As a robustness check, I instead estimate the first stage ranks of $\alpha_i \mid X_i^0$ directly from the conditional distribution of $X_i \mid X_i^0, Z_i$ using a kernel to nonparametrically condition on the prior X_i^0. The results are broadly similar to the results using the quantile regression based rearrangement method, but are noisier.

There are several considerations that make this rearrangement estimator more attractive than "directly" computing $\alpha_i = \frac{X_i - X_i^0}{S(Z_i) - X_i^0}$. First there are practical concerns related to estimation: since this expression involves dividing one random variable by another random variable that can cross zero, this fraction will be very noisy when $S(Z_i) - X_i^0$. And, since the α_i are individual specific quantities, each α_i is calculated using only one observation. Since division is nonlinear, any measurement error in the prior enters the denominator and does not disappear if observations are binned together, even asymptotically. For these reasons, estimating α_i directly for each i introduces some technical challenges. Further, as I discuss in Section 4, estimating the rank (or conditional rank) of α_i and using this as the first stage will be robust to some kinds of misspecification.

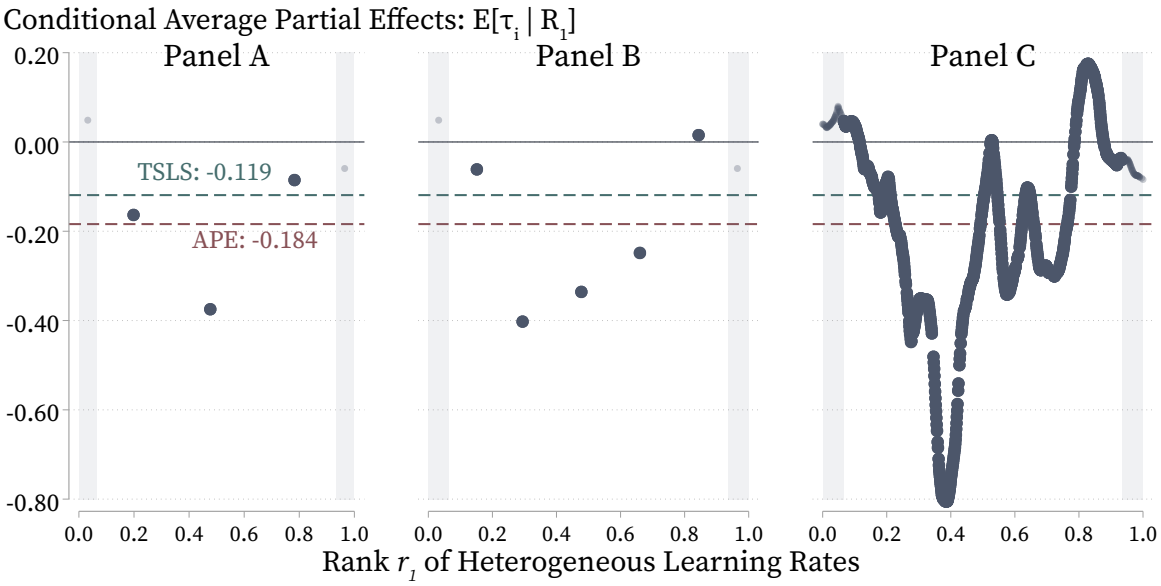
C.2. Non-Parametric Estimation of the Rank of the Learning Rate

Estimating the rank of the unobservable as in (32) requires the estimation of a conditional distribution function. In the empirical application in Section 5, I estimate 250 quantile regressions and use the rearrangement estimator proposed by Chernozhukov et al. (2010) to estimate the conditional ranks of the unobservable learning rate α_i. I specify the quantile regression as a regression of the shift in beliefs $X_i - X_i^0$ on the difference between the signal and the prior $S_i - X_i^0$, which recovers the rank of the learning rate α_i (conditional on the sign of the difference $S_i - X_i^0$). However, it is also possible to estimate the conditional density in (32) directly with a kernel.

I use an alternative non-parametric specification for the first stage estimation. Separately for each treatment arm, I use a kernel to estimate the CDF of the posterior conditional on a particular value of the prior. I estimate this conditional CDF at every value of the prior in the data and use the estimated CDF to recover the rank of the unobservable learning rate α_i. This non-parametric estimator is more demanding than

[19] While these regressions are particularly intuitive to interpret given the Bayesian model linear updating model, Masten and Torgovitsky (2014) show that the Chernozhukov et al. (2010) rearrangement method can be used whenever the endogenous regressor satisfies the necessary latent index condition in Masten and Torgovitsky (2014, 2016).

FIGURE C.1. Non-Parametric First-Stage Estimation of Ranks of α_i



Note: This figure replicates the results of Figure 1 in the main text using a non-parametric estimator for the conditional rank of α_i in the first stage. It plots estimates of conditional average partial effect $\mathbb{E}\left[\tau_i \mid F_{\alpha_i|W_i}(\alpha_i) = r_1\right]$ for different ranks r_1 of the learning rate α_i. Panel A and Panel B present results after collapsing to three and five equally-sized bins. Panel C displays all 400 underlying estimates. The shaded region denotes where the kernel begins to truncate due to the boundaries at 0 and 1. These regions are excluded when constructing the binned averages, instead the average estimates within these regions are plotted in light gray.

the quantile regression rearrangement estimator and as a result, observations in sparse regions of the prior distribution drop out. This leaves a sample of 2,056, relative to the main sample of 3,022. With these estimated ranks, I then estimate the second stage control function exactly as in Section 5. To make estimates comparable across designs, I re-estimate the TSLS coefficients on this subsample.

I replicate Figure 1 in the main text with this alternative specification and present the results in Figure C.1. The results are qualitatively similar to the main results, though both the APE estimate and the TSLS estimate are larger in magnitude. The shape of the conditional treatment effect curve is broadly similar, though the treatment effects are relatively smaller at low ranks of the conditional learning rate α_i. Thus, whereas the specification in the main text suggests that the treatment effects curve is broadly linear

in the rank of the learning rate α_i, this specification suggests that the treatment effects curve may be U-shaped. As I mentioned in the main text, conditional effects are harder to estimate for people with the lowest ranks of the learning rate, since these are people for whom the treatment has the smallest effect on beliefs (i.e., the first stage is weakest). This may explain why the shape of the curve in the lower ranks is most sensitive to the specification.

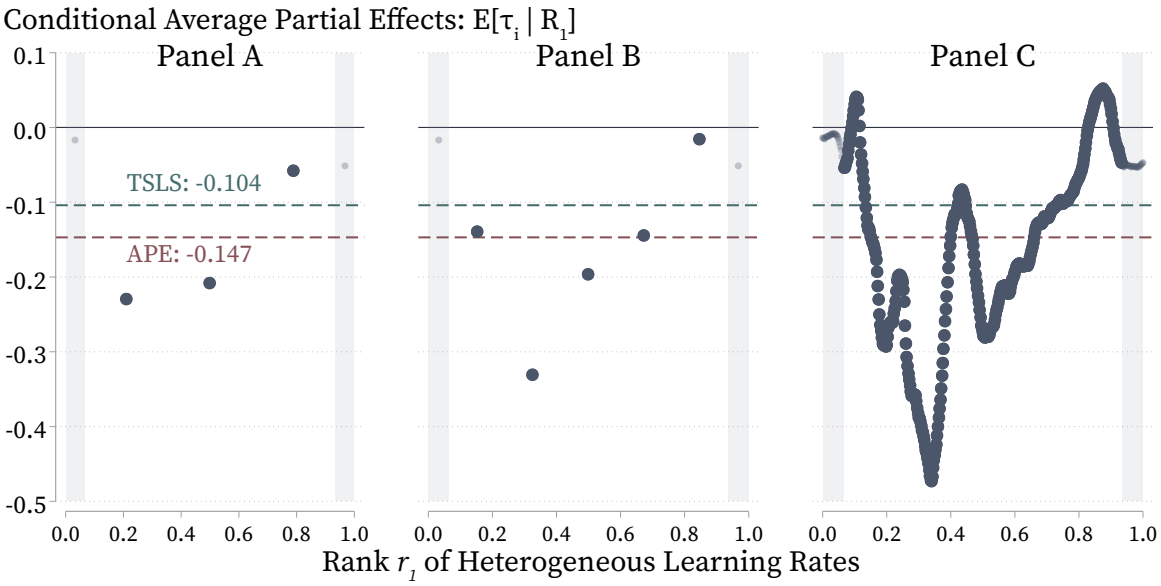
However, even with this alternative specification, the most general patterns are consistent. The average partial effect (-0.184) is over 50% larger in magnitude than the corresponding 2SLS estimate (-0.118). And, just as in the main specification, the effect is much larger in the bottom two-thirds (-0.235) of the learning rate distribution than in the top third (-0.077), or in the bottom eighty percent (-0.232) than in the top twenty percent (0.021). Across various specifications, I find that the treatment effects are quite small among the people with the highest ranks of the learning rate α_i, which are precise the people who receive the most weight in the 2SLS estimate. This is consistent with the fact that the 2SLS estimate is much smaller in magnitude than the APE estimate.

C.3. Alternative Controls for the Prior

If the linear updating model is correctly specified, then it is sufficient to control for the prior linearly, which eases estimation. It is possible to condition on the prior non-parametrically by using, for example, a two-dimensional kernel to condition on both $F(\alpha_i)$ and the prior and then regressing the outcome directly on the prior in the intermediate regression. This is a more data intensive approach that will use fewer observations for each intermediate regression.

In Figure C.2 I show that the results are nearly identical if I include a quadratic term for priors in the intermediate regression. This is perhaps unsurprising because the OLS estimation is already quite flexible, since this estimation happens only after conditioning non-parametrically on the rank of the learning rate α_i. In this sense, the linear control in the internal regression is comparable to a linear control with many flexible interactions with the learning rate α_i. Since this is already a quite flexible model, adding a quadratic term does not change the results very much. This suggests that using a linear control for the prior at this step in estimation is not overly restrictive. Since this reduces the dimensionality of the non-parametric step, this is attractive from an estimation perspective.

FIGURE C.2. Quadratic Control for Prior Beliefs



Note: This figure replicates the results of Figure 1 in the main text, adding a control for the prior squared into the "internal" regression that is estimated for each value of the control function. The TSLS estimate is also recomputed with this control. It plots estimates of conditional average partial effect $\mathbb{E}\left[\tau_i \mid F_{\alpha_i|W_i}(\alpha_i) = r_1\right]$ for different ranks r_1 of the learning rate α_i. Panel A and Panel B present results after collapsing to three and five equally-sized bins. Panel C displays all 400 underlying estimates. The shaded region denotes where the kernel begins to truncate due to the boundaries at 0 and 1. These regions are excluded when constructing the binned averages, instead the average estimates within these regions are plotted in light gray.

C.4. TSLS Estimates from Settele (2022) Are Similar Without Controls

Since I estimate the control function without additional controls, I also re-estimate the main specification TSLS without these controls to make these estimators more comparable. In Table C.1, I present results comparing the more parsimonious specification used in this paper with the original estimates from Settele (2022). I will then compare these estimates to the estimates of the APE. In Column 1, I replicate the specification in Table 5, Panel C of Settele (2022). In Column 2, I omit the weights used in the original paper. In Column 3, I also omit the covariates used in the original paper. For the sake of simplicity, I will omit the covariates used in the original paper in the control function estimation. Thus, the control function estimates are most comparable to the 2SLS estimates in

Column 3, and not the 2SLS estimates in Column 1 (and the original paper).

There is no conceptual reason that this estimation cannot be done conditional on covariates. However, since the control function estimation is a semi parametric estimator, including these covariates is technically demanding for estimation without any conceptual payoff and introduces new researcher degrees of freedom, making this exercise less transparent. For the purposes of an empirical illustration, I will use the most parsimonious specification possible.

Note that roughly half of the difference between the estimates in this application and in the original paper is due to reweighting in the original paper, and half is due to the use of controls. Omitting the weights but keeping the original controls increases the magnitude of the point estimate from –0.087 to –0.094. Omitting the controls then increases the magnitude further from –0.087 to –0.105.

Table C.1. 2SLS Estimates are Similar with and without Weights and Covariates

	Paper Specification	No Weights	No Covariates
Policy demand index	-0.087	-0.094	-0.105
	(0.038)	(0.038)	(0.041)
Introduce gender quotas	-0.085	-0.084	-0.095
	(0.054)	(0.054)	(0.056)
Statutory affirmative action	-0.171	-0.175	-0.189
	(0.053)	(0.052)	(0.055)
Stricter equal pay legislation	-0.177	-0.186	-0.195
	(0.053)	(0.053)	(0.055)
Wage transp. within companies	0.026	0.026	0.015
	(0.065)	(0.065)	(0.067)
Introduce reporting website	-0.144	-0.167	-0.185
	(0.092)	(0.088)	(0.092)
Increase subsidies to childcare	-0.009	-0.023	-0.032
	(0.053)	(0.052)	(0.055)

Notes: This table replicates the 2SLS estimates from Settele (2022). The first column replicates the specification in Table 5, Panel C of Settele (2022). The second column omits the weights used in the original paper. The third column also omits the covariates used in the original paper. N = 3,022 for all outcomes, except for the wage transparency policy (2,003) and the reporting website policy (1,019). Standard errors for the 2SLS estimates are heteroskedasticity-robust.

C.5. Alternative Choices of the Bandwidth

There is a relatively arbitrary choice of bandwidth in the main specification. We need to choose a bandwidth to approximate conditioning on the continuous rank of the learning rate α_i. In the main specification, I use a rule of thumb bandwidth, which is gives a half-width of approximately 0.0635. In this section, I will present results for a range of bandwidths $h \in \{0.01, 0.025, 0.05, 0.075, 0.10\}$. The maximum and minimum differ by an order of magnitude, but estimates are qualitatively similar across all bandwidths. The APE estimate at the largest bandwidth $h = 0.1$ is –0.142 and at the smallest bandwidth $h = 0.01$ is –0.152, which suggests that results are not particularly sensitive to the choice of bandwidth, at least within the range of bandwidths that I consider.

Table C.2. APE Estimates Are Not Sensitive to Choice of Bandwidth

	TSLS	APE (Various Bandwidths)				
		$h = 0.01$	$h = 0.025$	$h = 0.05$	$h = 0.075$	$h = 0.10$
Policy demand index	-0.105	-0.152	-0.156	-0.153	-0.147	-0.142
	(0.041)	(0.099)	(0.041)	(0.040)	(0.038)	(0.036)

Note: This table presents APE estimates using different values of the bandwidth, as well as the corresponding TSLS estimate as a benchmark. Standard errors for the APE are computed by bootstrapping the two-stage procedure 100 times. I use an Epanechnikov kernel and evaluate the control function at 400 grid points on the interval [0, 1].